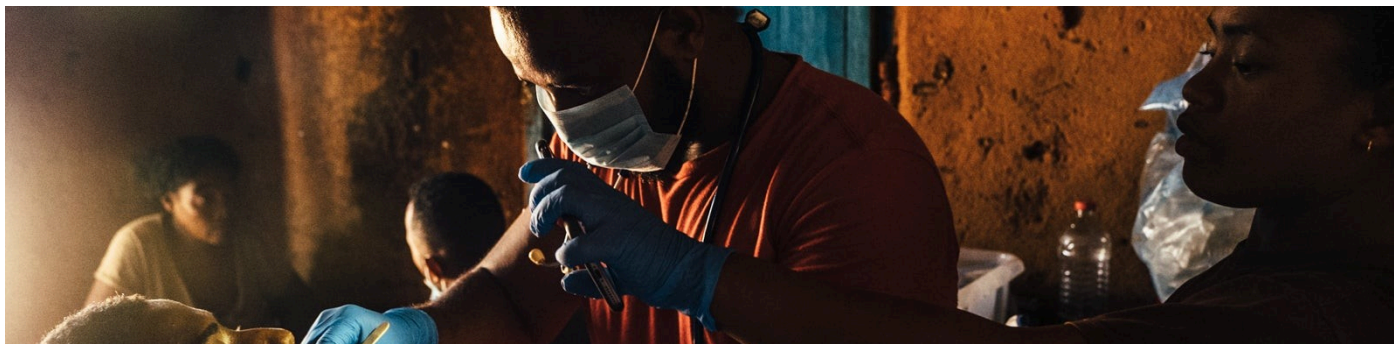


20 | OVERVIEW



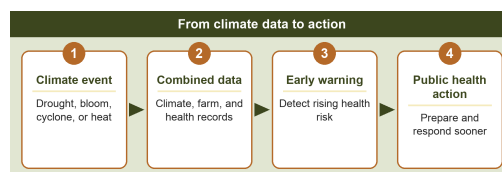
Climate-Smart Public Health: Connecting Environment and Well-Being

Abstract

Public health data and climate data have usually been kept in separate systems, making it hard to predict how environmental changes will affect people's health. Harvard Chan researchers led by Christopher Golden are building an integrated data platform in Madagascar that combines climate, environmental, agricultural, and health information. The goal is early warning systems that help health officials act before climate-related illness strikes.

Introduction

When a drought hits, crops may fail, and months later children may become malnourished. When warm seas cause harmful algal blooms, people who eat contaminated seafood can be poisoned. When a cyclone strikes, it can cause food shortages and lasting mental health effects. These chains of cause and effect connect the environment to human health, but they are hard to see if climate data and health data never meet. In Madagascar, researchers are building a system that brings these data together. They call the approach Climate-Smart Public Health.



How Do Scientists Study This?

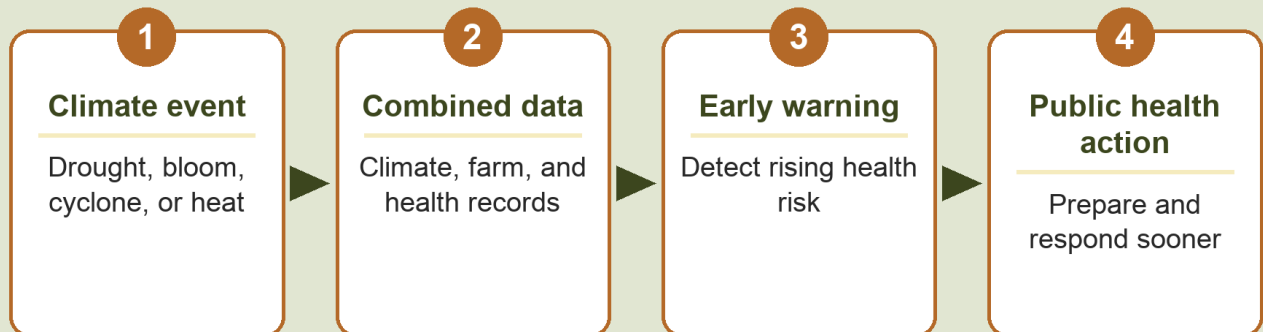


FROM THE ARTICLE

Project locations: Madagascar and Nepal. Map: Harvard Chan C-CHANGE.

The team is building an integrated research data platform that combines climate, environmental, agricultural, and health data in one place. Using advanced data science methods, researchers look for connections between climate stressors and health outcomes. The work is led by the Harvard T.H. Chan School of Public Health and the Harvard Data Science Initiative, in partnership with four Madagascar government ministries: Public Health, Environment, Agriculture, and Fisheries and the Blue Economy. The researchers aim to connect the platform to DHIS-2, a widely used health information system, and other health surveillance data, so the tools can be easily adapted and shared with other countries.

From climate data to action



Study the Process

Combine data

Climate, environmental, agricultural, and health records are brought together in one platform.

Find connections

Data science methods reveal how climate stressors are linked to specific health outcomes.

Warn early

Understanding these links allows early warning systems and faster public health responses.

What Did They Discover?

The platform supports several kinds of analysis. Researchers can monitor droughts to assess the risk of crop failures and malnutrition, and detect harmful algal blooms to watch for marine food poisoning. They can track how cyclones affect food insecurity and mental health, and measure how climate fluctuations affect respiratory conditions like asthma. They can evaluate how disease patterns change across seasons and climate zones, and assess how temperature, rainfall, and soil moisture affect the spread of diseases carried by insects. These insights give decision makers critical information to anticipate, prepare for, and respond to climate-related health risks. The work has already begun to expand to Nepal.

Climate-Smart Public Health

4

Madagascar ministries partnering on the platform

6

example climate-health links the platform examines

2

countries so far: Madagascar and Nepal

Because the tools are designed to connect with common health surveillance systems, they can be adapted for regions around the world with similar challenges.

20 | TAKEAWAYS

Takeaways

Connect the data

Linking climate and health data reveals hidden paths from environment to illness.

Predict, then prevent

Early warnings help officials act before droughts or blooms cause harm.

Partner with governments

Four Madagascar ministries help turn research into real public health action.

Key Terms

Integrated data platform

A system that combines different types of data, such as climate and health records, so they can be analyzed together.

Early warning system

A tool that detects signs of an approaching hazard so people can prepare before harm occurs.

Harmful algal bloom

A rapid growth of algae that produces toxins, which can contaminate seafood and water.

Vector-borne disease

An illness spread by insects or other animals, such as malaria carried by mosquitoes.

Health surveillance

The ongoing collection and analysis of health data to detect and respond to health problems.

Activity

Build an Early Warning

Students design an early warning rule that uses climate data to predict a health risk.

Materials: Scenario cards: drought, algal bloom, cyclone, heavy rain; Sample data table provided by the teacher; Poster paper

1. Choose a scenario card and draw the chain from the climate event to the health outcome.
2. Decide which data you would need to watch, such as rainfall, sea temperature, or crop yields.
3. Write an early warning rule, for example: if rainfall stays below a set level for two months, alert nutrition programs.
4. Name which of the four government ministries would need to act on your warning.
5. Explain why combining climate and health data makes your rule possible.

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Climate-Smart Public Health.
<https://hsph.harvard.edu/research/climate-health-c-change/research/climate-smart-public-health/>